7 A German astronomer, Johannes Kepler, deduced that for a planet in a circular orbit around the Sun, its period of rotation T and the radius of its orbit r, is related by

$$T^2 \propto r^3 .$$

(a) Using Newton's law of gravitation and considering the mass of Saturn to be M, show that, for a circular orbit of a moon around Saturn,

$$T^2 = \frac{4\pi^2}{GM} r^3 .$$

[3]

(b) Table 7.1 contains some of the data for the major moons of Saturn.

Table 7.1

moon	period T / 10^6 s	mean distance from centre of Saturn r / 10^9 m	lg (T / s)	lg (r / m)
Enceladus	0.121	0.238	5.08	8.38
Tethys	0.164	0.295	5.21	8.47
Rhea	0.380	0.501		
Titan	1.38	1.26		
Lapetus	6.83	3.56	6.83	9.55

Complete Table 7.1 for the moons Rhea and Titan. [2]

(c) Fig. 7.1 shows a graph representing the variation of lg (T / s) with lg (r / m) for the moons of Saturn, with some of the data from Table 7.1 plotted.

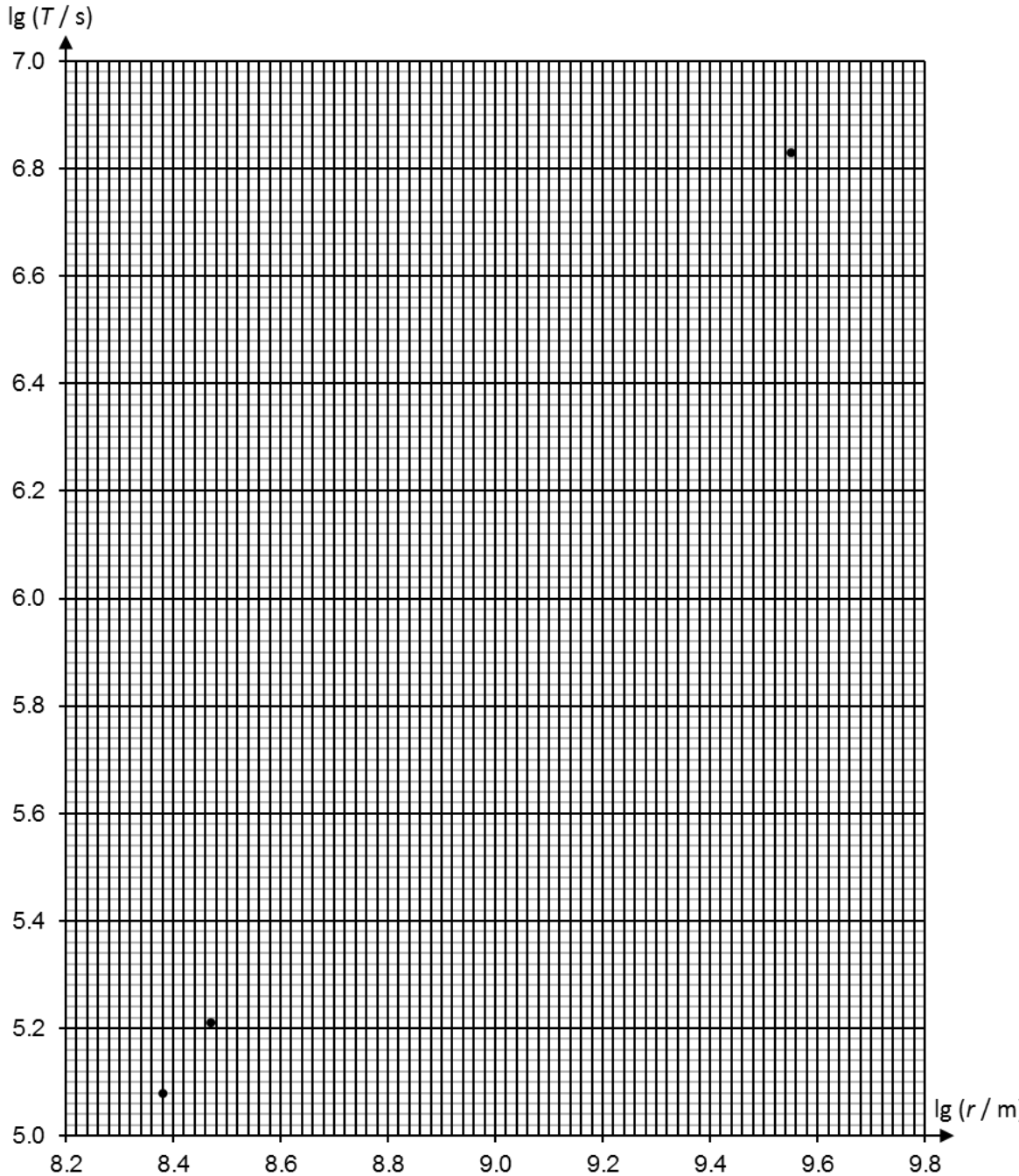

Fig 7.1

On Fig 7.1,

(i) plot the points corresponding to the moons Rhea and Titan. [1]

(ii) draw the line of best fit for all the data points. [1]

(d) **(i)** Determine the gradient of the graph in Fig 7.1.

gradient = …… [2]

(ii) Hence, discuss whether the data in Fig 7.1 supports the relation given in **(a)**.

……

……

…… [2]

(e) Dione, which is another moon of Saturn, has an orbital radius of 3.78×10^5 km.

Using the graph in Fig. 7.1, determine the period of Dione's orbit around Saturn.

period = …….. s [2]

(f) Scientist were able to determine the mass of planets in the Solar System through studying the orbits of their moons.

(i) Using Fig 7.1, determine the mass of Saturn.

mass = ……. kg [2]

(ii) A student studying the orbits of the moons of Saturn decides to determine the mass of Saturn with the orbital radius and period of Titan only.

Discuss one disadvantage of using this method as compared to **(f)(i)**.

.....

.....

..... [2]

(iii) It was reported that another moon of Saturn, Mimas, has a period of rotation 7.78×10^4 s and orbits at a height of 1.79×10^5 km above Saturn's surface. Comment on the accuracy of this report.

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..... [3]